

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456344.67062 +/- 0.0005 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.154 +/- 0.048
Transit depth (Rp/Rs)^2	2.38 +/- 1.49 [%]
Orbital Inclination (inc)	79.22 +/- 2.91
Airmass coefficient 1 (a1)	3.69 +/- 2.52
Airmass coefficient 2 (a2)	-0.66 +/- 0.29
Scatter in the residuals of the lightcurve fit is	79.51 %
Best Comparison Star	None
Optimal Aperture	6.7
Optimal Annulus	10.91
Transit Duration (day)	0.034 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.154 ± 0.048 vs 0.1368 (deviation 0.3σ)
Duration fit vs published	0.034 d vs 0.0758 d (deviation 1.14σ)
Mid-time O–C	0.6 min
Depth SNR	2.7
Residual scatter	79.51 %

Light curve

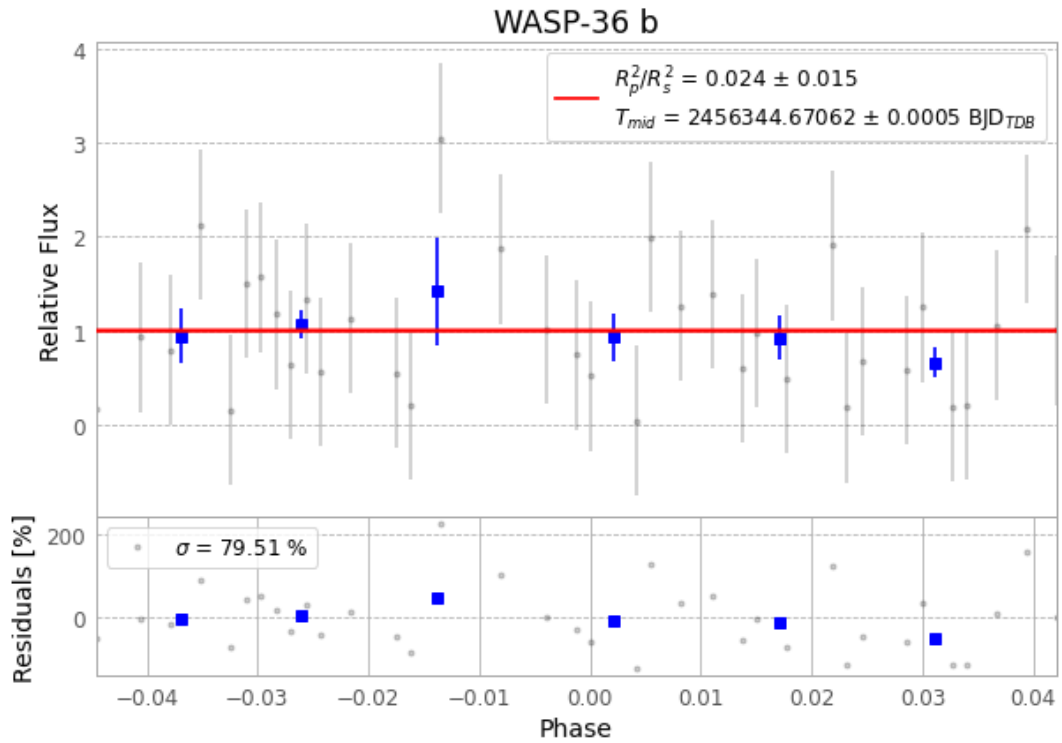


Figure 1 — FinalLightCurve_WASP-36 b_2013-02-20.png (SHA-256 29163175a5ba71a6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [173, 239], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4afede749e2085ce...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

66 FITS frames (43 MB), WASP-36130221021512.FITS → WASP-36130221053012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-130221.log (1881c6e3bdd6...), FinalLightCurve_WASP-36 b_2013-02-20.png (29163175a5ba...), FinalLightCurve_WASP-36 b_2013-02-20.csv (5fb5a9d138fc...)

Compute resources

Wall time 164.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-21 00:10Z.